Technext

TECHNICAL REPORT 03

EnergieBee core Multi-country handling

Markets, currencies, regions and tariffs

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Content is reproduced from the source document `docs/multi-country-handling.md`, which remains the editable original.

REPORT OVERVIEW

Nothing in the core knows which country it is in. Everything country-shaped is a row in a table, and everything time-shaped is a setting on the home.

The pieces

Piece	Table	What it holds
Market	`market`	Country, time zone, currency, price interval, meter interval, default slot length, bidding zone
Currency	`currency`, `site.currency`, `tariff.currency`	GBP and EUR in minor units (pence, cents). A tariff in a different currency from its home is rejected.
Regions	`region`, `region_alias`, `site_region`	Every way a country is divided, with aliases between numberings, versioned per home
Taxes	`tax_rule`	Ordered rules per country: percent, per kWh, or per period (can be negative)
Tariffs	`tariff`, `rate_interval`, windows, calendars	Flat, time-of-use, dynamic, French off-peak and Tempo, fixed and capacity charges
Providers	`market_provider`	Which adapter fills which role in which market
Slot length	`site.slot_s`	15, 30 or 60 minutes per home
Labels	`origin` columns	sourced, provider, user, placeholder or synthetic

Example: three homes, same code

	UK home	French home	Dutch home
Market	GB	FR	NL
Currency	GBP (pence)	EUR (cents)	EUR (cents)
Region	One of 14 distribution areas	National, plus the meter's own off-peak hours	Bidding zone
Tariff shape	Flat, day/night, half-hourly dynamic	Off-peak windows, Tempo colour days, monthly fee	15-minute spot-indexed, yearly credit
Default slot	30 min	30 min	15 min
Slots on the short clock-change day	46	46	92
Tax	VAT (sourced)	Placeholder values	Placeholder values
Real data source today	Octopus, Sunsynk, Tariffscanner	None, synthetic demo only	None, synthetic demo only

Example: one place, several codes

Regions are not one list. A place can carry a code in each scheme, and the core links them with aliases.

Postcode	Octopus letter	MPAN distributor id	Region name
BB12	G	16	North West England

- The 14 GB distribution areas exist in three numberings: Octopus letters, MPAN ids and Tariffscanner ids and names. Tariffscanner's numbering is 1 to 14 and is not the MPAN numbering, so it is never assumed equal.
- NESO carbon regions are a separate scheme and are never aliased to distribution areas, because they divide the country differently. The same postcode is in distribution area G and carbon region 3.
- France prices energy nationally, so a French home needs a market and a per-meter off-peak window, not a region price.

Example: taxes stacked in order

Rules apply in order, so a percent rule is charged on top of earlier ones. Illustrative, with placeholder values:

Order	Rule	Kind	Effect
1	Energy tax	per kWh	Added to the unit price
2	VAT	percent	Charged on the price including the energy tax
3	Yearly credit	per period, negative	Spread by slot length over the year

French and Dutch values are placeholders until checked against official sources. A home with no tax rules for its country is taxed at 0 and labelled placeholder.

How a home is processed

1. The home's market sets the clock, currency, default slot and bidding zone.
2. The registry lists the providers for that market. Roles with no provider are shown as empty.
3. Fetched data is stored as facts with their original units, then converted to minor units.
4. The build prices each slot with the tariff in force and the market's tax rules, in the home's currency.
5. Every figure carries its origin, so placeholder and synthetic values are always visible.

Adding a country

1. Add a `market` row: time zone, currency, price and meter intervals, default slot, bidding zone.
2. Add its tax rules from official sources, with dates.
3. Add its region schemes and aliases.
4. Add its calendars if the tariffs need them (for example French Tempo colours).
5. Add `market_provider` rows for each role, switched on only when the adapter exists and the licence is confirmed.
6. Write adapters for what is missing: a price source, a meter reader, a carbon feed.

Steps 1 to 5 are data. Only step 6 is code, and each adapter is a contained job that writes into the existing tables.

What is not done

- No real data source for France or the Netherlands yet. Linky, P1, ENTSO-E and tounify are listed but off.
- French and Dutch taxes are unverified placeholders.
- Only GB has regions and a catalogue in the database. Other countries are stand-in rows.
- Dutch net metering ends on 1 January 2027, so per-slot export prices for 2026 are not modelled.

See `docs/what-it-does.md` for how providers are chosen and `docs/capabilities-and-roadmap.md` for the remaining work.